The University of Edinburgh

College of Science and Engineering

Mathematics 3 Honours

MATH10003 Financial Mathematics

Saturday, 12th December 2015

14:30 – 16:30

Chairman of Examiners – Professor J M Figueroa-O'Farrill

External Examiner – Professor G A Young

Credit will be given for the best **THREE** answers

Calculators and other electronic aids.

A scientific calculator is permitted in this examination.

It must not be a graphical calculator.

It must not be able to communicate with any other device.

This examination will be marked anonymously.

(1) Assume that today's stock price, S, is £30 and it is expected either to increase by 19% or to decrease by 16% every four months over the next two four-month periods (two-period Binomial tree, $n = 0, 1, 2$). Also, suppose that the continuously compounded risk-free rate of interest is 5% per annum.

(a) Verify the absence of arbitrage and compute the prices (at period $n = 0$ and $n = 1$, i.e. $t = 0$ and $t = 4$ months) of the 8-month European Call option Φ with strike price $K = £29$, written on the non-dividend paying stock S.

[10 marks]

(b) Compute the prices (at $n = 0$ and $n = 1$) of the 8-month American call option Φ with strike price $K = £29$, written on the non-dividend paying stock S.

[7 marks]

(c) Recall that the binomial tree aims at being an approximation of the real evolution of the price of a stock. Does this 2-period binomial model reproduce a specific volatility σ of the real observed market price of a stock? Explain.

[3 marks]

(d) In relation to the 2-period binomial model described above compute the following.

(d.1) The fair delivery price K at time $t = 0$ and $t =$"four-months" of a forward contract with maturity in *eight*-months and written on the underlying S.

[3 marks]

(d.2) The fair delivery price K at time $t = 0$ of a forward contract with maturity in *four*-months and written on the underlying S.

[2 marks]

(2) (a) Take the one-period binomial model with maturity $N = T = 1 = \Delta t$, a stock $(S_n)_{n=0,1}$ and a bank account (paying continuously with rate r) with accumulation factors $(B_n)_{n=0,1}$. The involved parameters for the stock are the usual ones u, d, p_u, p_d and assume that the market is arbitrage free.

(a.1) Given the portfolio $h = (x, y)$, where x denotes a cash deposit of x and y the number of stocks held, write down the equation for its market value $(V_n^h)_{n=0,1}$.
Moreover, write the possible outcomes of V_n^h explicitly for $n = 0$ and $n = 1$ depending on the value of the stock at $n = 1$.

[4 marks]

(a.2) Let $\Delta \in \mathbb{R}$ and take the portfolio $h := (0, \Delta)$. An agent holds at time $t = 0$ the portfolio $h = (0, \Delta)$ and a short position in one European call option Φ written on S_1. Denote the price of this European call option at $t = 0$ by c_0 and by, $(I_n^\Delta)_{n=0,1}$, the value of the investments.
Write an expression for I_n^Δ at $n = 0$ and $n = 1$.
Deduce the Δ that makes the value of I_1^Δ constant with relation to the stock value, in other words, the value of Δ such that $I_1^\Delta(S_1(u)) = I_1^\Delta(S_1(d))$.

[5 marks]

(a.3) Assume that Δ in the conditions of the previous question exists and has been computed.
State the *Law of one price.*
Using the law of one price, compare the cashflow of $(I_n^\Delta)_{n=0,1}$ with another conveniently chosen cashflow and deduce an expression determining explicitly c_0 as a function of the remaining problem parameters (Δ, S_0, r, Δt and I_1^Δ).

[6 marks]

(b) A loan is to be repaid by an annuity payable annually in arrears. The annuity starts at a rate of £300 per annum and increases each year by £30 per annum. The annuity is paid for 20 years. Repayments are calculated using a rate of interest of 7% per annum effective.
Calculate:

(b.1) The amount of the loan.

[4 marks]

(b.2) The capital outstanding immediately after the 5th payment has been made.

[3 marks]

(b.3) The capital and interest components of the final payment.

[3 marks]

(3) (a) Let $T \in [0,\infty)$ and let $(W_t)_{t\in[0,T]}$ be a standard Brownian motion.

State Itô's isometry over the time interval $[0,T]$. State any assumption you might need.

Use Itô's isometry along with the formula $2ab = (a+b)^2 - a^2 - b^2$ to show that

$$\mathbb{E}\left[\left(\int_0^T X_s \mathrm{d}W_s\right)\left(\int_0^T Y_s \mathrm{d}W_s\right)\right] = \mathbb{E}\left[\int_0^T X_s Y_s \mathrm{d}s\right].$$

[8 marks]

(b) Let $T \in (0,\infty)$. Take a general market with a stock $(S(t))_{t\in[0,T]}$ and a bank account paying interest continuously in time at rate r. Assume the market to be free of arbitrage.

Take a European Call and Put, written on S, maturing at time T and with the same strike $K > 0$. Denote by $C(t, S_t; T, K)$ and by $P(t, S_t; T, K)$ the price at time $t \in [0,T]$ for a stock value of S_t of the call and put respectively, maturing at T and with strike K.

Prove the put-call parity formula at time $t = 0$ that states:

$$S_0 + P(0, S_0; T, K) = C(0, S_0; T, K) + Ke^{-rT}.$$

[8 marks]

(c) Suppose that a Portuguese company is to issue a bond with 6 years to maturity. The bond has an annual coupon of £100 per year for the next 6 years in coupon interest. The interest rate is assumed to be 5% (annual compounding). At the end of the 6th year the company will pay £500 to the bond's owner.

Calculate the *present* value of the bond. Calculate the *future* value, at the end of the 6h year, of this bond.

[9 marks]

(4) Consider a stock with price process $(S_t)_{t\geq 0}$ satisfying the following stochastic differential equation (SDE),

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \qquad S_0 \in (0, \infty)$$

where $(W_t)_{t\geq 0}$ is a standard Brownian motion, $-\infty < \mu < +\infty$ is the drift and $\sigma > 0$ is the diffusion coefficient.

(a) By using Itô's formula, show that the solution of the above SDE is given by the geometric Brownian motion S

$$S_t = S_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma W_t}, \quad t \in [0, T].$$

[8 marks]

(b) Let the evolution of stock price $(S_t)_{t\in[0,T]}$ be given by a geometric Brownian motion with drift r and volatility σ.

State the distribution and parameters of the random variable $\ln(S_T)$, then prove that $e^{-rT}\mathbb{E}[S_T \mathbb{1}_{S_T > K}] = S_0 N(d_+)$, where d_+ is given by

$$d_+ = \frac{\ln\left(\frac{S_0}{K}\right) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}},$$

and N denotes the distribution function of a standard Normal variate.

[8 marks]

(c) Take $T \in (0, \infty)$ and assume a Black-Scholes market, where the evolution of stock price $(S_t)_{t\in[0,T]}$ is given by a geometric Brownian motion with drift r and volatility σ and the bank account pays interest continuously at rate r.

In a chooser option, the holder can choose whether the option is a call or a put after a specified time, say $T_1 < T$. Suppose that the chosen call or put options are defined on the same non-dividend paying stock S with same maturity T and strike price $K > 0$.

The pay-off of the chooser option at time T_1 is given by

$$V(T_1, S_{T_1}) = \max\Big\{\ C(T_1, S_{T_1}; T, K)\ ;\ P(T_1, S_{T_1}; T, K)\ \Big\},$$

where $C(T_1, S_{T_1}; T, K)$ and $P(T_1, S_{T_1}; T, K)$ are the prices of a call and a put option respectively at time T_1 for a current stock price of S_{T_1}, with the maturity T and strike price K.

Show that the price of the chooser option at current time $t = 0$ is

$$V(0, S_0) = S_0 N(d_+) - Ke^{-rT} N(d_+ - \sigma\sqrt{T}) + Ke^{-rT} N(-y + \sigma\sqrt{T_1}) - S_0 N(-y)$$

where

$$d_+ = \frac{\ln\left(\frac{S_0}{K}\right) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}, \qquad y = \frac{\ln\left(\frac{S_0}{K}\right) + rT + \frac{\sigma^2}{2}T_1}{\sigma\sqrt{T_1}},$$

and N denotes the cumulative distribution function of a standard Normal variate.

Suggestion: Use the Put-Call parity relationship. Recall the expressions for the prices of European calls and puts in this market (strike K, maturity T and written on S)

$$C(0, S_0; T, K) = S_0 N\big(d_+\big) - Ke^{-rT} N\big(d_+ - \sigma\sqrt{T}\big)$$
$$P(0, S_0; T, K) = Ke^{-rT} N\big(-d_+ + \sigma\sqrt{T}\big) - S_0 N\big(-d_+\big)$$

where d_+ is given as above, T is the maturity and K is strike price of the option. N denotes the distribution function of a standard Normal variate.

[9 marks]

[End of Paper]